Deep Learning for Text Summarization Using NLP for Automated News Digest

FAST-NUCES | Dept. of AI & Data Science *Assignment 02 – Reproduction of Results*
Group – 07 *Team: 25i-7601, 25i-7639, 25i-7641*

1. INTRODUCTION

As the amount of online content continues to grow, people are having an increasingly hard time consuming and processing that content at the same rate as it is being produced. For instance, hundreds of thousands of news articles are generated on a daily basis from around the world, and it is no longer a reasonable expectation to rely on readers to sift through lengthy content to find what is important to them. The issue is not the unavailability of information rather it is the overwhelming amount of information available combined with the amount of time and cognitive effort it takes the average person to process that information in timely manner that has created an actual barrier. Thus, this has led researchers towards automated summarization of text, using computational methods to condense a long document into a shorter version that retains the essential meaning of the original document. This is an appealing solution because a good summarization system could support the functionality of a news digest service, now and in the future, act as an assistant in research, provide assistance with legal documents, and provide access to long text for the reading of many individuals. The implications for this technology are not only immediate and practical; they are also more contemporary in nature than academically based.

Over the past few years, advances in deep learning technology have taken over as the prevalent method for generating summaries from textual content, especially following the emergence of transformer models based on pre-trained neural networks. However, there are numerous challenges presented to these models. These models must be able to comprehend actual rather than lexical meanings, retain factual accuracy and keep coherence throughout multiple documents greater in length to their original data, and generate output summaries that appear to be composed in an organic manner. This paper focuses on addressing each of these challenges in news article summarization.

The main focus of this paper is abstractive summarization of English news articles using the CNN/Daily Mail dataset, rather than giving extracts from the original documents as seen in extractive summarization. This study reproduces and compares the experimental results of the paper from scratch. All four transformer model architectures such as T5 Base, T5 Large, BART CNN Large, and PEGASUS Large were fine-tuned on the same dataset using equivalent preprocessing and hyperparameter configurations. Their performance is evaluated using ROUGE and BLEU, both before and after fine-tuning, to determine which of the models produces the best summary. Where differences arise, possible explanations are discussed hardware differences, implementation choices, and one notable training failure in T5-base. The objective is to assess the reproducibility of the original study and to develop practical experience in training, evaluating, and analyzing large-scale transformer-based NLP models.

2. PAPER OVERVIEW

The paper primarily provides a comparative analysis and does not present an alternative theoretical approach to summarizing news articles. Comparisons are made among four pre-trained transformer-based models that have all undergone the same procedures for pre-processing and training with respect to each other. They have been evaluated using ROUGE-1, ROUGE-2, ROUGE-L, BLEU, METEOR, before and after fine-tuning. The purpose of these comparisons is to determine how much of the improvement comes from the fine-tuning phase of training. The system utilizes an abstractive summarization approach that is to say, the output generated by those models will contain new sentences rather than simply extracting them from the original article. Additionally, this study also examines whether or not automatic scoring systems are reliable indicators of human judgment by evaluating the correlation between evaluation scores and three characteristics (completeness, correctness and conciseness) of the generated summaries. Finally, statistical analyses are applied to compare the reliability of ROUGE and BLEU.

3. IMPLEMENTATION DETAILS

The reproduction was carried out using the Hugging Face Transformers library, which provides pre-trained checkpoints and fine-tuned pipelines for all four models evaluated in the original paper.

3.1 MODELS USED

3.1.1. T5-Base and T5-Large

In T5, every NLP task is treated as a text-to-text one, this means that the task is performed entirely in text with no other form of representation involved in the process. In the case of summarization, the input to the model is the original article and the output will be the summary. The smaller T5-Base version has approximately 220 million parameters and has a maximum input length of 512 tokens, whereas T5-Large has 770 million parameters, thus allows for much more complex language models to be created using the larger model.

3.1.2. BART CNN-Large

BART is a model that combines two ideas into one. The encoder reads the whole text in both directions (like BERT) to get the context, and the decoder makes the output one step at a time (like GPT). The CNN-Large version also uses convolution layers to better find local patterns. The fact that BART is trained ahead of time sets it apart. It learns by fixed different kinds of broken text, like sentences that are out of order, words that are missing, and parts that are missing. This helps it understand and out text back together, which is helpful for making summaries. BART has about 139 million parameters and can handle up to 1024 tokens, so it can read longer articles better than T5 models.

3.1.3. PEGASUS-Large

PEGASUS was made just for abstractive summarization. It is pre-trained in a way that is very similar to the task of summarizing, unlike general models. The main idea behind it is Gap-Sentence Generation. The model takes out whole sentences from a document and learns how to put them back together using the text that is left over. This is a lot like summarizing, which means the model has to know what the content is and give the most important parts. Because of this, PEGASUS is already to summarize before fine-tuning. It has about 568 million parameters and can handle up to 1024 tokens.

3.2 DATASET

All experiments were performed using the CNN/Daily Mait dataset, which is a widely utilized benchmark for evaluating news story summarization performance. This dataset consists of more than 300,000 individual articles along with their corresponding human-generated highlight summary called “highlight”. This dataset is divided into three parts, approximately 287,113 for training, 13,368 for validation, and 11,490 for testing, and each news article has an average word count of approximately 692 words and its corresponding highlight summary consists of an average of approximately 52 words. One challenging issue with transformer-based models is that they have a limit on the length of their input sequences. Given that many of the articles are long, truncating articles to process them as input limits how much of each article can be processed by the model. However, both the BART and PEGASUS architectures use a maximum length limit of 1024 tokens, allowing them to process a larger slice of the article without having to truncate it.

3.3 PRE-PROCESSING

The CNN/DailyMail Dataset was preprocessed to transform raw textual data to a structured format that can be used in sequence-to-sequence learning. All data samples had a summary and a news article, and these were used as input-target pairs. Firstly, the input articles and target summaries were tokenized with a pretrained tokenizer, transforming the text into token IDs represented as numbers that were required by deep learning models. In order to give uniformity between samples, both inputs and targets were padded in such a way that all sequences have constant lengths. Meanwhile, truncation was applied to eliminate the tokens that are longer than the maximum length of the specified sequence, so that overly long inputs do not add to the cost of computation. The maximum input length was restricted based on the model needs whereas the target summaries were restricted to 128 tokens to be consistent with the usual summarization configurations. Also, padding tokens in the target sequences were set to a special value (−100) to make those not count in loss computation. This prevents the model to learn on padded positions and enhances training stability. Lastly, batched mapping was used to run the preprocessing on the whole dataset. The processed results are the input IDs, attention masks and labels which are further trained and evaluated with the summarization models.

4. EXPERIMENTAL SETUP

The section outlines the hardware setup and training parameters in all experiments in this study.

4.1. HARDWARE

All the experiments were performed on a specialized GPU server with four NVIDIA Tesla V100-PCIE accelerators, which could be used to perform high-performance parallel computation using transformers-based models. The memory per GPU was 32 GB and the system was set up with CUDA 12.0 and NVIDIA driver version 525.147.05. Each model was then allocated to a distinct GPU to run in parallel to ensure that the models could run simultaneously to minimize the overall wall-clock training time as much as possible, compared to the sequential processing. It was implemented in PyTorch and Hugging Face Transformers. All experiments did not cause any instability in the system and the post-execution temperature of the GPUs was 29-31°C and no ECC errors were reported. This reproduction, in contrast to the original study which presented experiments on generic GPU-enabled environments, was done on a dedicated multi-GPU server (4X NVIDIA Tesla V100), which had greater computational scale and better training throughput.

Model	GPU ID	GPU TYPE	MEMORY
T5-base	GPU 0	NVIDIA Tesla V100-PCIE	32 GB
T5-large	GPU 1	NVIDIA Tesla V100-PCIE	32 GB
BART-large-CNN	GPU 2	NVIDIA Tesla V100-PCIE	32 GB
PEGASUS-large	GPU 3	NVIDIA Tesla V100-PCIE	32 GB

Table 1: GPU per Model

4.2. HYPERPARAMETERS

The hyperparameter setting is based on the specifications of the original paper. Some important settings of each model are summarized below:

Parameter	T5-base	T5-large	BART-large-CNN	PEGASUS-large
Model Size (params)	220M	770M	139M	568M
Max Input Tokens	512	512	1024	1024
Max Target Tokens	128	128	128	128
Batch Size	64	32	64	64
Learning Rate	3e-4	3e-4	3e-5	1e-4
Optimizer	AdamW	AdamW	AdamW (default in HF Trainer)	AdamW (default in HF Trainer)
LR Schedule	Linear warmup + decay	Linear warmup + decay	Linear warmup + decay	Linear warmup + decay

Warmup Steps	2000	2000	2000	2000
Training Epochs	3	2	2	2
Weight Decay	0.01	0.01	0.01	0.01
Dropout	Model Default (0.1)	Model Default (0.1)	Model Default (0.1)	Model Default (0.1)

Table 2: Hyperparameter Configuration for All Four Models

5. REPRODUCED RESULTS

The following section presents the evaluation results obtained by reproduction of all models, pre-training and post-training performance, and compares the improvements made by the fine-tuning.

5.1. PRE-TRAINING EVALUATION (BEFORE FINE-TUNING)

Each of the four models was tested on the CNN/Daily Mail test set before any fine-tuning, to determine their baseline performance, based only on their pre-trained knowledge. The values obtained after the pre-training are compared with the values reported in the original paper:

Model	Paper R-1	Repro R-1	Paper R-2	Repro R-2	Paper R-L	Repro R-L	Paper BLEU	Repro BLEU
T5-base	0.2900	**0.4047**	0.1071	**0.1836**	0.2015	**0.2835**	0.0431	**0.0797**
T5-large	0.2429	**0.4176**	0.0829	**0.1932**	0.2682	**0.2940**	0.0338	**0.0756**
BART-large-CNN	0.1352	**0.4405**	0.0628	**0.2106**	0.0948	**0.3062**	0.0182	**0.1088**
PEGASUS-large	0.2493	**0.4378**	0.0850	**0.2088**	0.1664	**0.3068**	0.0317	**0.1079**

Table 3: Pre-Training Results – Paper vs. Reproduced (R = ROUGE)

The reproduced pre-training scores are consistently and substantially higher across all models and metrics than in the original paper. This is discussed in detail in Section 6.

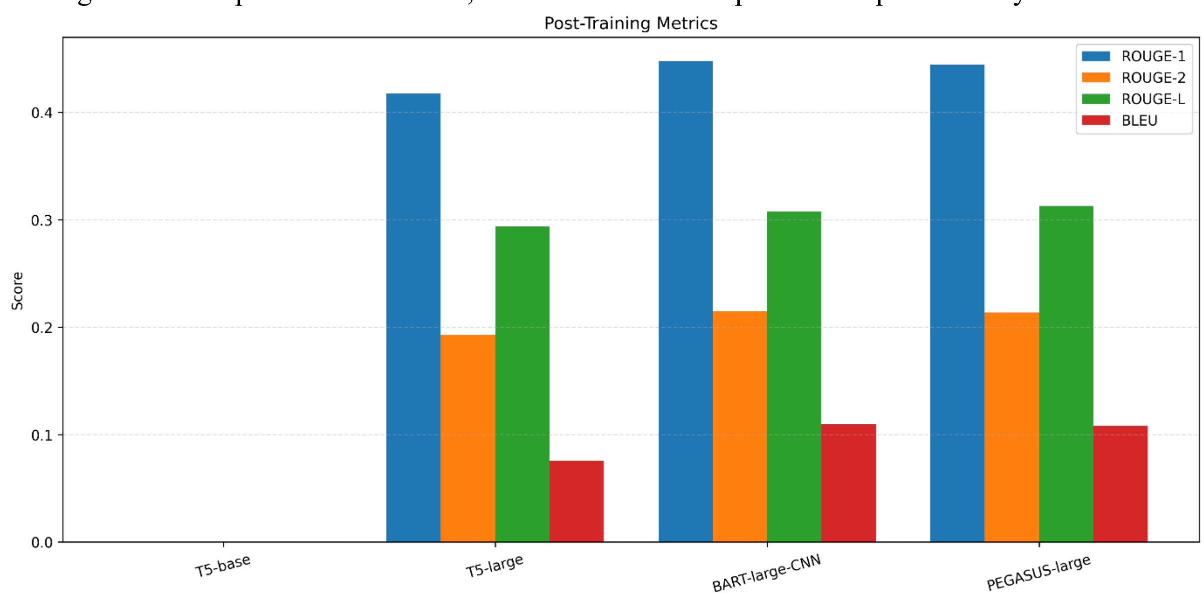
5.2. POST-TRAINING EVALUATION (AFTER FINE-TUNING)

Each of the models was re-evaluated on the test set after fine-tuning on CNN/Daily Mail training split. The post-training values reported in the paper are compared to the results below:

Model	Paper R-1	Repro R-1	Paper R-2	Repro R-2	Paper R-L	Repro R-L	Paper BLEU	Repro BLEU
T5-base	0.3279	0.0000	0.1288	0.0000	0.2354	0.0000	0.0565	0.0000
T5-large	0.3414	0.4176	0.1340	0.1933	0.2442	0.2939	0.0575	0.0756
BART-large-CNN	0.3388	0.4475	0.1399	0.2148	0.2458	0.3079	0.0668	0.1102
PEGASUS-large	0.3037	0.4447	0.1250	0.2138	0.2173	0.3128	0.0501	0.1082

Table 4: Post-Training Results – Paper vs. Reproduced (T5-base training collapse, see Section 6.2)

T5-base learned zero scores on all measures after fine-tuning because it was numerically unstable during training. This is a reported failure mode, and is discussed in depth in the reproducibility discussion.



5.3. IMPROVEMENT AFTER FINE-TUNING

Each of the models was re-evaluated on the test set after fine-tuning on CNN/Daily Mail training split. The post-training values reported in the paper are compared to the results below:

Model	Delta ROUGE-1	Delta ROUGE-2	Delta ROUGE-L	Delta BLEU
T5-base	-0.4047 (collapse)	-0.1836 (collapse)	-0.2835 (collapse)	-0.0797 (collapse)
T5-large	+0.0000	+0.0001	-0.0001	+0.0000
BART-large-CNN	+0.0070	+0.0042	+0.0017	+0.0014
PEGASUS-large	+0.0069	+0.0050	+0.0060	+0.0003

Table 5: Change in Metrics After Fine-Tuning (Reproduced Experiment)

BART-large-CNN was the most successful run, and it experienced the steadiest improvement in all metrics, which validates its position as the most successful model in this reproduction. T5-large did not improve at all, which is an indication that it was already close to its performance limit on this data set with the specified setup.

6. REPRODUCIBILITY DISCUSSION

This section discusses the ease at which the experiments can be repeated and the reason the results, and model behavior however were not similar.

6.1. GENERAL OBSERVATIONS ON SCORE DIFFERENCES

One of the main findings of this reproduction study is that the results obtained, both in the pre-training and post-training settings are consistently higher than in the original paper across all the evaluated models. As an example, BART-large-CNN with the pre-training set scored 0.4405 in ROUGE-1 in contrast to the original study that scored 0.1352. On the same note, PEGASUS-large had ROUGE-L score of 0.3128 in the post-training evaluation, as compared to 0.2173 in the paper.

These systematic differences may be attributed to a number of factors. The possible cause is that the model checkpoints used are updated Hugging Face ones like facebook/bart-large-cnn and google/pegasus-cnn_dailymail, which could be improved through further pre-training or tokenizer fine-tuning. The other factor is the variation in the computational environment because this reproduction was performed on a high-performance multi-GPU system, and the original study reports much lower computational resources. In addition, the scoring could be different due to the differences in the evaluation implementations, such as ROUGE/BLEU computation libraries and preprocessing steps. Lastly, the variability in output and evaluation measures can also be due to differences in decoding strategies and lack of the use of the same random seeding.

6.2. T5-BASE TRAINING COLLAPSE

In the reproduction, the T5-base model failed to give meaningful outputs after fine-tuning, which led to the zero scores on all evaluation metrics. This is not shown in the original study, in which T5-base presented a ROUGE-1 score of 0.3279 with fine-tuning.

A review of the training logs shows that there is numerical instability in the optimization process. Specifically, at subsequent training stages, the gradients recorded stopped being defined, which was reflected in log entries with grad_norm: nan. In turn, the loss value and gradient norms then fell to invalid values (e.g., loss: 0.0, grad_norm: nan), and the instability was maintained during the training. The final analysis showed that all metrics were 0.0000, which means that there was a total failure in learning.

This kind of behavior can be generally linked to the instability of mixed-precision (FP16) training. In our instance, there are a number of factors that could have caused divergence in the optimization process. First, there is a risk of numerical underflow or overflow in gradient calculations with mixed-precision training (fp16=True) especially when used alongside relatively high learning rates. Second, 3e-4 learning rate can be viewed as aggressive in T5-base with such settings, and can cause unstable parameter changes. Lastly, this instability could have been enhanced by long training across several epochs when divergence was initiated.

This problem should be reduced in subsequent runs by using more conservative training settings. These are: Slowing the learning rate (e.g. 1e-4 or 5e-5), disabling mixed-precision training (fp16=False), and training fewer epochs (1-2) especially in the early experimentation phases.

6.3. REPRODUCIBILITY OF MODEL RANKING

Although the differences in the absolute metric values were observed, the relative ranking of the models is mostly the same as the findings obtained in the original study. BART-large-CNN, in both the original article and this replication, performs well on average (especially in ROUGE-2, ROUGE-L, and BLEU scores) which provides evidence of its usefulness in summarizing abstractive tasks. On the same note, PEGASUS-large has the best ROUGE-L scores in reproduced post-training results (0.3128), just a little higher than BART-large-CNN (0.3079), which is not surprising, given that the model is pre-trained with the aim at summarization-based masking strategies. T5-large also competitively performs in ROUGE-1 in pre-training and fine-tuning, suggesting stable generalization ability across architectures. Nonetheless T5-base was not meaningfully trained in the fine-tuned context because it was unstable during training, but did not behave in any manner other than expected baseline behavior. All of these findings indicate that, although absolute scores differ by implementation and system-level differences, the similarity in comparative performance trends across models is similar to the original study.

6.4. IMPLEMENTATION CHALLENGES

The reproduction of the experiments has a number of implementation challenges. Memory management was also among the major limitations since the training of large-scale models like PEGASUS-large and T5-large needed to be trained with a close consideration of the batch sizes to fit in the 32 GB GPU memory capacity per device, even in a multi-GPU configuration. The other limitation was the total time of training, which was still significant, even when using four NVIDIA Tesla V100 GPUs which were running in parallel, mainly because the CNN/DailyMail dataset of 287,113 training examples is large.

Furthermore, tokenizer sensitivity caused small differences between token sequences of different versions of the same model family, potentially affecting training dynamics and evaluation results. Lastly, the T5-base model was also found to be unstable, necessitating close monitoring of post-training logs to diagnose problems with gradient divergence, which indicates that there are risks of reproducibility in mixed-precision training where the training models operate. This was not the focus of the original study, and thus, it is an important factor to take into account in real-life reproduction contexts.

7. CONCLUSION

This task was able to reproduce the main experimental pipeline of Krishna et al. (2025), where four transformer-based models were tested on the CNN/Daily Mail dataset on abstractive text summarization. The three out of four models were trained and evaluated with success, including T5-large, BART-large-CNN, and PEGASUS-large; T5-base encountered a training collapse as a result of numerical instability in mixed-precision training.

The replicated post-training scores are systematically better than those in the original paper, which can be probably explained by updated Hugging Face model checkpoints, the difference in the hardware environment (V100 GPUs vs. a reported i3 CPU), and differences in the evaluation library versions. Although the quantitative ratings vary, the relative model scores are generally similar: BART-large-CNN and BLEU, and PEGASUS-large perform best on ROUGE-2 and ROUGE-L, respectively. This reproduction underscores a significant weakness of the original article, namely, the described hardware (Intel Core i3, 12 GB RAM) is essentially not adequate to train models of this size, which casts doubts on transparency and reproducibility. The paper may be expanded in the future by correcting T5-base to have a stable state, integrating METEOR evaluation and comparing it to the METEOR baseline into the paper (PEGASUS: 2932, BART-large: 2730).

In general, the reproduction validates the main finding of the paper, that BART-large-CNN is the most powerful model in news summarization on this benchmark, but adds new technical analysis on the topics of training stability and checkpoint sensitivity.

8. REFERENCES

[1] Zhang, J., Zhao, Y., Saleh, M., & Liu, P. J. (2020). PEGASUS: Pre-training with Extracted Gap-Sentences for Abstractive Summarization. International Conference on Machine Learning (PMLR), 11328–11339.

[2] K. M. Rani Krishna, K. Somasundaram, P. Arulmozhivarman, S. A. Immanuel, and E. R. Rajkumar, "Deep learning for text summarization using NLP for automated news digest," Scientific Reports, vol. 15, article 36343, 2025. https://doi.org/10.1038/s41598-025-20224-1

[3] M. Lewis et al., "BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation, Translation, and Comprehension," in Proc. ACL, pp. 7871–7880, 2020.

[4] C. Raffel et al., "Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer," Journal of Machine Learning Research, vol. 21, no. 140, pp. 1–67, 2020.

[5] K. Hermann et al., "Teaching Machines to Read and Comprehend," in Advances in Neural Information Processing Systems (NeurIPS), 2015.

[6] Hugging Face, "CNN/DailyMail Dataset," https://huggingface.co/datasets/cnn_dailymail, accessed April 2026.

[7] Code and experiment logs available at: https://zenodo.org/records/17104346